TensorFlow Environment Setup

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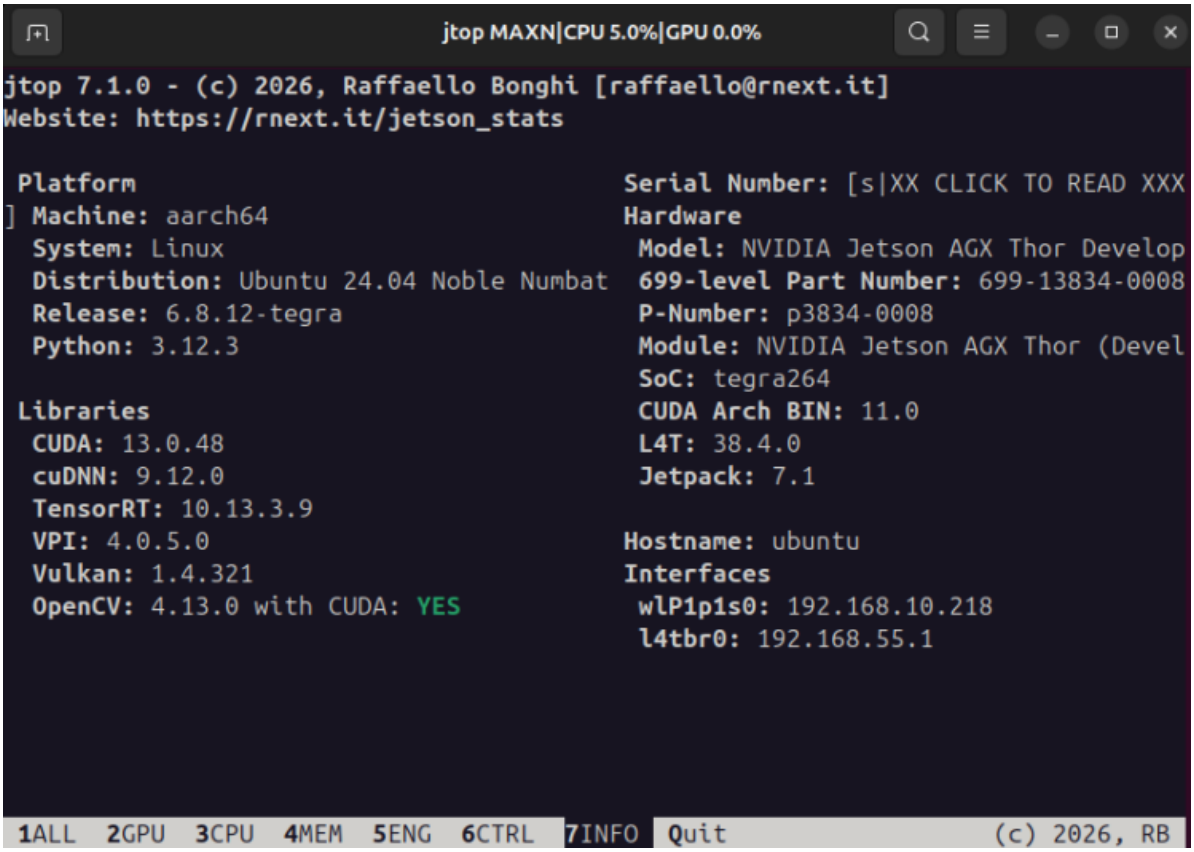
TensorFlow is an end-to-end machine learning and deep learning framework developed and open-sourced by Google, used for building and training various types of machine learning models.

1. System Information



Since the Jetson AGX Thor system uses Jetpack 7.1 and CUDA 13, and there is currently no official TensorFlow GPU installation package supporting the aarch64 architecture, only the CPU version of TensorFlow can be installed at present.

2. Resolving Dependencies

Because the system's built-in typing_extensions version is too low, it will cause TensorFlow installation to fail. Therefore, you need to upgrade the typing_extensions version first.

```
sudo pip3 install -U typing_extensions --ignore-installed --break-system-packages
```

```
jetson@ubuntu:~$ sudo pip3 install -U typing_extensions --ignore-installed --bre
ak-system-packages
Collecting typing_extensions
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions
Successfully installed typing_extensions-4.15.0
WARNING: Running pip as the 'root' user can result in broken permissions and con
flicting behaviour with the system package manager. It is recommended to use a v
irtual environment instead: https://pip.pypa.io/warnings/venv
jetson@ubuntu:~$
```

3. Install TensorFlow

Enter the following command to install the CPU version of TensorFlow.

```
sudo pip3 install tensorflow --break-system-packages
```

```
jetson@ubuntu:~$ sudo pip3 install tensorflow --break-system-packages
Collecting tensorflow
  Using cached tensorflow-2.21.0-cp312-cp312-manylinux_2_27_aarch64.whl.metadata
 (4.4 kB)
Collecting absl-py>=1.0.0 (from tensorflow)
  Using cached absl_py-2.4.0-py3-none-any.whl.metadata (3.3 kB)
Collecting astunparse>=1.6.0 (from tensorflow)
  Using cached astunparse-1.6.3-py2.py3-none-any.whl.metadata (4.4 kB)
Requirement already satisfied: flatbuffers>=25.9.23 in /usr/local/lib/python3.12
/dist-packages (from tensorflow) (25.12.19)
```

4. Verify installation

```
python3 -c "import tensorflow as tf; print(f'TensorFlow: {tf.__version__}')"
```

```
jetson@ubuntu:~$ python3 -c "import tensorflow as tf; print(f'TensorFlow: {tf.__
version__}')"
TensorFlow: 2.21.0
jetson@ubuntu:~$
```